Experiment 5

AIM: Jenkins is an open-source automation server used for implementing Continuous Integration (CI) and Continuous Deployment (CD) pipelines. It helps automate the process of building, testing, and deploying applications, thereby improving development efficiency and software quality.

Theory: Creating Pipeline in Jenkins for Testing and Deployment

A Jenkins Pipeline is a suite of automated processes that defines how software is built, tested, and deployed. It enables Continuous Integration (CI) and Continuous Deployment (CD) by automating each stage of the software development lifecycle.

1. Overview of Jenkins Pipeline

A pipeline in Jenkins is defined using a script called a **Jenkinsfile**, which contains a series of steps organized into stages. Each stage represents a phase in the lifecycle such as building, testing, and deployment.

Pipelines help in:

- Automating workflows
- Reducing manual intervention
- Ensuring consistent and repeatable processes

2. Types of Jenkins Pipelines

- **Declarative Pipeline**

  - Uses a structured and simplified syntax
  - Easier to read and maintain

- **Scripted Pipeline**

  - Uses Groovy scripting
  - Provides more flexibility and control

3. Pipeline Structure

A typical Jenkins pipeline consists of the following components:

- **Pipeline Block**: Defines the entire pipeline
- **Agent**: Specifies where the pipeline will run
- **Stages**: Logical divisions of the pipeline
- **Steps**: Individual tasks within each stage

4. Stages in CI/CD Pipeline

1. Source Stage

- Retrieves code from version control systems like Git
- Ensures latest code is used for build

2. Build Stage

- Compiles source code
- Installs dependencies
- Generates build artifacts

3. Test Stage

- Executes automated test cases
- Validates functionality of the application
- Helps in early detection of bugs

4. Deploy Stage

- Deploys the application to a server or container
- Can be done on local server, cloud, or Docker container Ensures application is available for end users

5. Testing in Jenkins Pipeline

Testing is an essential part of the pipeline:

- Unit testing verifies individual components
- Integration testing checks interaction between modules
- Automated tests ensure code quality and reliability Test results are displayed in Jenkins dashboard

6. Deployment in Jenkins Pipeline

Deployment automates the release of applications:

- Can deploy to staging or production environments
- Uses tools like Docker or cloud platforms
- Ensures faster and error-free delivery

7. Advantages of Pipeline Automation

- Continuous feedback on code changes
- Faster development and deployment cycles
- Reduced human errors
- Improved software quality
- Easy monitoring and debugging

Conclusion

Thus, creating a pipeline in Jenkins enables automation of building, testing, and deployment processes. It plays a crucial role in implementing CI/CD practices, ensuring faster, reliable, and efficient software delivery.

Jenkins / KC03

Status
Changes
Workspace
Build with Parameters
Configure
Delete Project
Rename

Builds >

No builds

Project KC03

This build requires parameters:

Fname

City

Bandra

Build

Cancel

Next

Jenkins 2.541.1

Jenkins / KC01

Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename

Builds >

No builds

KC01

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